

Palpitations

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Palpitations are the awareness of heartbeats. The sensation may feel like pounding, fluttering, racing, or skipping beats. Other symptoms—for example, [chest discomfort](#) or [shortness of breath](#)—may be present depending on the cause of the palpitations.

Palpitations are common. Some people find them unpleasant and alarming, but they rarely indicate a life-threatening heart disorder. Many people without heart disease also have palpitations.

Causes of Palpitations

Ordinarily, people do not notice the beating of their heart. However, many people can feel their heart beating when something causes it to beat more forcefully or rapidly than usual. Such rapid, forceful beats are usually a normal response by the heart (sinus tachycardia). Causes include the following:

- Exercise
- Strong emotions (such as anxiety, fear, or pain)
- Low blood count ([anemia](#))
- [Low blood pressure](#)
- [Fever](#)
- [Dehydration](#)

In other cases, palpitations result from a disturbance of heart rhythm (arrhythmia).

Types of arrhythmias

[Arrhythmias](#) range from harmless to life threatening.

The **most common arrhythmias** include

- [Atrial premature beats](#)
- [Ventricular premature beats](#)

Both of these arrhythmias usually occur in people without a heart disorder and are harmless. The premature beat itself is not felt. What is felt is the following normal heartbeat, which occurs after a slight delay and is slightly stronger than usual. Although people feel as if their heart skipped a beat, it actually did not.

Other arrhythmias that cause palpitations and may be caused by serious heart disorders include

- [Paroxysmal supraventricular tachycardia](#) (PSVT)
- [Atrial fibrillation](#) or [atrial flutter](#)
- [Ventricular tachycardia](#)

These other arrhythmias involve the heart beating much faster than normal. Arrhythmias in which the heart beats too slowly rarely cause palpitations, although some people do feel the slow rate.

Causes of arrhythmias

Some arrhythmias (for example, atrial premature beats, ventricular premature beats, or PSVT) often occur in people who have no serious underlying disorders. Others are often caused by a serious heart disorder or a disorder elsewhere in the body.

Serious heart disorders include [angina](#), [heart attack](#) (acute coronary syndrome), congenital heart diseases (for example, the Brugada syndrome or congenital [long QT syndrome](#)), [disorders of heart valves](#), and conduction system disturbances (for example, [Wolff-Parkinson-White syndrome](#)).

Disorders not related to the heart that may cause arrhythmias include

- Overactive thyroid gland ([hyperthyroidism](#))
- Low level of oxygen in the blood (hypoxia)
- Low level of potassium in the blood ([hypokalemia](#))
- Low level of magnesium in the blood ([hypomagnesemia](#))
- Certain medications, illicit drugs, or substances, including alcohol, caffeine, [digoxin](#), nicotine, and some stimulants (such as [albuterol](#), amphetamines, cocaine, [epinephrine](#), [ephedrine](#), and [theophylline](#))
- High levels of hormones produced by a tumor of the adrenal glands ([pheochromocytoma](#))

Complications

Arrhythmias that cause the heart to beat too fast can cause complications (particularly in older adults). If the heart goes too fast, it may not be able to pump blood adequately, and people may feel faint or pass out or develop [heart failure](#). Heart failure occurs mainly in people who have previously had heart failure or a heart attack, although it can occur in other people if the heart rate is very fast or if the heart beats too fast for a long time. A rapid heart rate also increases the oxygen needs of the heart muscle. People who have narrowing of the arteries to the heart muscle ([coronary artery disease](#)) can develop chest pain due to angina or a heart attack (which may be fatal).

Some arrhythmias, particularly ventricular tachycardia, are unstable and can lead directly to [cardiac arrest](#).

Evaluation of Palpitations

Although not all palpitations are caused by a heart disorder, the consequences of some heart disorders are so serious that people with palpitations should usually be evaluated by a doctor. The following information can help people decide when to see a doctor and help them know what to expect during the evaluation.

Warning signs

In people with palpitations, certain symptoms and characteristics are cause for concern. They include

- Light-headedness or fainting
- Chest pain or pressure
- Shortness of breath
- Pulse rate over 120 per minute or less than 45 per minute
- Having heart disease or a family history of sudden death, recurrent fainting, or unexplained seizure disorder
- Symptoms occur during exercise, particularly if they cause loss of consciousness

When to see a doctor

People who have palpitations and any warning signs should go to an emergency department right away, as should those who have continuous, ongoing palpitations.

Those who have fainting, chest pain, or shortness of breath should call emergency services.

People without warning signs who have occasional palpitations or had an episode that stopped should call their doctor. The doctor will determine how quickly they need to be seen based on their age, underlying conditions, and other symptoms.

What the doctor does

Doctors first ask questions about the person's symptoms and medical history. Doctors then do a physical examination. What they find during the history and physical examination helps determine the possible cause.

Doctors ask

- How often palpitations occur
- How long palpitations last
- What factors (for example, emotional distress, activity, or intake of caffeine or other drugs) trigger or worsen palpitations
- What drugs or substances, including caffeine, the person is taking

Sometimes doctors ask the person to tap out the rate and cadence of palpitations because the pattern of beats can help them determine the cause.

The physical examination begins with the doctor checking the vital signs (pulse, temperature, and blood pressure). The pulse rate and whether the pulse is regular or irregular help suggest causes. An elevated temperature suggests fever is the cause. A very low blood pressure does not suggest a cause but indicates extreme urgency.

Doctors listen to the heart for abnormal sounds that might indicate a heart valve disorder or heart inflammation. They listen to the lungs for sounds that indicate heart failure. They look at and feel the front of the neck to see if the thyroid gland is enlarged or inflamed.

Palpitations that occur with other symptoms, such as shortness of breath, chest pain, weakness, fatigue, or fainting, are more likely to result from an abnormal heart rhythm or a serious disorder.



Some Causes and Features of Palpitations

Feature*	Possible Cause
Perception of an occasional skipped heartbeat	Atrial premature beats or ventricular premature beats
Sudden episodes of rapid heartbeats that abruptly slow to the normal rate; often in people who have had previous episodes of rapid heartbeats	PSVT or Wolff-Parkinson-White syndrome Less often ventricular tachycardia or congenital long QT syndrome
A constant sensation of rapid, irregular heartbeats	Atrial fibrillation
Fainting during palpitations	Ventricular tachycardia, inherited cardiomyopathies (including hypertrophic cardiomyopathy and arrhythmogenic right ventricular cardiomyopathy), atrial fibrillation, cardiac channelopathies (including long QT syndrome and Brugada syndrome)
Palpitations during exercise or an emotional situation	Sinus tachycardia (the heart's normal response to stress), particularly common in healthy people Abnormal heart rhythm caused by a decrease in blood flow triggered by exercise, more common in people who have coronary artery disease or congenital heart diseases
In people with a family history of fainting or sudden death	Cardiac channelopathies (including Brugada syndrome and congenital long QT syndrome), or inherited arrhythmogenic cardiomyopathies, including

